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EEG Data Augmentation Method for Identity Recognition Based on Spatial–Temporal Generating Adversarial Network

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Abstract: Traditional identity recognition methods are facing significant security challenges due to their vulnerability to leakage and forgery. Brainprint recognition, a novel biometric identification technology leveraging EEG signals, has emerged as a promising alternative owing to its advantages such as resistance to coercion, non-forgeability, and revocability. Nevertheless, the scarcity of high-quality electroencephalogram (EEG) data limits the performance of brainprint recognition systems, necessitating the use of shallow models that may not perform optimally in real-world scenarios. Data augmentation has been demonstrated as an effective solution to address this issue. However, EEG data encompass diverse features, including temporal, frequency, and spatial components, posing a crucial challenge in preserving these features during augmentation. This paper proposes an end-to-end EEG data augmentation method based on a spatial–temporal generative adversarial network (STGAN) framework. Within the discriminator, a temporal feature encoder and a spatial feature encoder were parallelly devised. These encoders effectively captured global dependencies across channels and time of EEG data, respectively, leveraging a self-attention mechanism. This approach enhances the data generation capabilities of the GAN, thereby improving the quality and diversity of the augmented EEG data. The identity recognition experiments were conducted on the BCI-IV2A dataset, and Fréchet inception distance (FID) was employed to evaluate data quality. The proposed method was validated across three deep learning models: EEGNET, ShallowConvNet, and DeepConvNet. Experimental results indicated that data generated by STGAN outperform DCGAN and RGAN in terms of data quality, and the identity recognition accuracies on the three networks were improved by 2.49%, 2.59% and 1.14%, respectively.

Keywords: data argumentation; deep learning; biometric identification; generating adversarial network (GAN); electroencephalogram (EEG)

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1. Introduction

With the rapid advancement of information technology, identity recognition has become a critical aspect of modern societal security systems. However, traditional methods like passwords and fingerprints frequently encounter security vulnerabilities such as susceptibility to leakage and forgery [1–5]. Thus, there is an urgent need to explore more secure and reliable technologies for identity recognition. Brainprint recognition [6], an emerging biometric identification technology, has garnered considerable attention among researchers. Based on the individual differences in brainwave signals, brainprint recognition technology achieves precise identification of individuals through the collection, processing, and analysis of electroencephalogram (EEG) signals. This technology not only boasts high accuracy and reliability [7–10], but also remains imperceptible and tamper-resistant from external manipulation [11,12]. Furthermore, it is cancelable [13–16], which is a unique advantage that traditional biological features do not possess. It holds substantial promise in enhancing security applications, and has aroused the interest of researchers.

Unlike data types such as images, speech, or natural language, which have access to abundant data resources, collecting EEG signals is particularly complex and expensive [17].

Due to the specialized nature of measuring EEG signals, specific acquisition equipment is required, and measurements must be conducted under particular environmental and conditional settings. This process involves not only collecting data from different subjects but also entails high costs associated with acquisition, data analysis, and annotation. Variations in data collection configurations across different experimental setups hinder the integration of collected data into a unified large-scale dataset. Moreover, the data collection process requires subjects to maintain a high level of concentration, and prolonged sessions can induce fatigue, thereby compromising data accuracy and potentially invalidating the data [18,19]. Furthermore, considering the privacy concerns associated with EEG data, such as age and gender [20], emotions [21], cognitive workload [22], cognitive and memory performance [23], and health status [24], the large-scale collection of EEG data is typically not feasible. As a result, EEG datasets used in practical applications are often limited in scale. Complex features require a large number of samples for accurate modeling; having limited data restricts the training effectiveness of models. Moreover, models tend to overfit on small datasets, becoming overly reliant on noise in the training data, which results in a lack of generalization ability for unknown or variable signals. Therefore, this limitation has become a significant factor constraining the advancement of research in EEG-based biometric recognition.

Data augmentation (DA) technology offers an effective strategy to address the challenge posed by the limited EEG dataset [25–27]. Data augmentation refers to the process of learning and transforming original EEG data, generating new samples that retain similar features while introducing subtle variations. This approach effectively expands the dataset size and enhances data diversity without the need for additional data collection efforts. Through data augmentation, various types of data variations can be simulated, such as shifts in time, frequency, amplitude, or noise levels. These augmented datasets enable machine learning models to learn from a broader range of examples, improving their ability to generalize to unseen EEG signals. By enhancing dataset diversity through data augmentation, researchers can mitigate the limitations imposed by small-scale EEG datasets, thereby facilitating more robust and effective training of EEG-based models for various applications [26,28,29].

In brainwave recognition, EEG signals are presented in the form of multi-channel time-series data, which exhibit features across time, spatial, and frequency domains [24,30,31]. Given the complexity of these time-series formats, conventional data augmentation methods such as cropping, rotating, or noise addition exhibit limitations. Cropping may result in the loss of critical information, while rotating EEG signals can disrupt their temporal features and meaningful patterns, thereby affecting the model's ability to make accurate judgments. Furthermore, excessive or inappropriate noise addition may obscure important features within the EEG signals, causing the model to fail to capture the real information during the learning process. Consequently, these methods are often unsuitable for augmenting EEG data [26]. A generative adversarial network (GAN) [32] is a type of generative model designed to create new data by modeling the distribution of original data. In recent years, GANs have gained significant attention in the field of data generation and have achieved notable success in various applications, including image generation [33], style transfer [34], text-to-image synthesis [35], and more.

GANs have been employed for EEG data generation, showcasing their potential in this domain [36–40]. Some studies utilize CNN-based GANs for EEG data augmentation [41,42]. In addition, other research integrates recurrent neural networks (RNNs) into GAN frameworks for data augmentation [36,43], focusing specifically on the temporal features inherent in EEG data. The locality of convolution operations makes it difficult for CNNs to capture long-range feature dependencies, which limits their ability to capture global features [44–46]. The RNN is designed to capture temporal features in sequential data. Due to its sensitivity to the input order and its step-by-step processing characteristics, it is not well suited for capturing static spatial features [47], Additionally, it faces issues such as vanishing gradients over long sequences [48]. To date, generating data from limited EEG datasets remains challenging due to the scarcity and complex spatiotemporal features of EEG data. Preserving internal features and semantic information during data augmentation presents a critical challenge that requires careful consideration and innovative solutions.

This paper introduces a novel self-attention mechanism inspired by the transformer encoder [49], embedded within a GAN framework featuring both temporal and spatial encoders. The model was designed to capture temporal and channel dependencies inherent in EEG data, preserving its complex spatiotemporal features during data generation. To address the challenge of data scarcity, the approach avoided directly fitting the distribution of original data during generative model training; it integrated the generator's output with the original data, aiming to minimize the distribution gap between original and generated data. This strategy accelerated model convergence and prevented discriminator overfitting. The proposed method effectively generated realistic EEG data and enhanced the performance of classification models.

The main contributions are as follows:

(1) This paper introduces a novel approach for multi-channel EEG data augmentation that was directly performed in the original signal space, by eliminating the data transformation and feature extraction stage, this approach helps to prevent potential errors or information loss, preserving the authenticity and complexity of the EEG signals.
(2) This paper presents a novel EEG data augmentation model that was specifically designed to capture and preserve the spatiotemporal features of EEG signals. By leveraging self-attention mechanisms to capture global dependencies across channels and time, the model enabled more effective feature extraction and integration, ultimately resulting in higher-quality augmented data that better reflect the complexity of real-world EEG signals.
(3) This paper evaluates the quality of generated data by computing the Fréchet inception distance (FID) [50] metric and conducting principal component analysis (PCA) visualization. The comparison encompassed state-of-the-art models such as RGAN and DCGAN. To further demonstrate effectiveness, several classification models were trained using the augmented data. The results showed that the augmented data significantly enhanced the accuracy of these classification models.

2. Related Work

2.1. GAN for EEG Data Augmentation

In 2014, Goodfellow et al. first proposed the generative adversarial network (GAN), which consists of a generator and a discriminator and demonstrates powerful generative capabilities. Since then, GANs have sparked considerable research interest, resulting in numerous derivative generative networks. While there has been limited research on using GANs to augment EEG data, existing studies indicate its feasibility as a method for EEG data augmentation.

Luo et al. [41] proposed a conditional Wasserstein GAN (CWGAN) framework for generating realistic EEG data in the form of Differential Entropy (DE). They evaluated this approach on two publicly available EEG datasets used for emotion recognition, SEED [21] and DEAP [51]. Experimental results showed that using CWGAN to generate EEG signals significantly improved the accuracy of emotion recognition models. Zhang et al. [27] employed short-time Fourier transform (STFT) to convert time-series signals into spectrogram images. They subsequently conducted an evaluation and comparison of various data augmentation techniques, including deep convolutional generative adversarial networks (DCGAN) [52], geometric transformations, autoencoders (AE) [53], and variational autoencoders (VAE) [54]. They utilized convolutional neural networks to classify motor imagery signals and compared the classification performance after data augmentation. The results indicated that DCGAN outperformed other methods in enhancing classification performance. Sudhanva [55] proposed a Wasserstein GAN with a gradient penalty (WGAN-GP) model: they firstly extracted nine descriptive features from raw data, and then trained WGAN-GP on these extracted features and utilized the trained model to generate a new set of synthetic data feature. Zhao et al. used preprocessed EEG topographic maps in identity recognition scenarios, and generated training data with FastGAN-ASP [56].

In the aforementioned studies, the data were initially subjected to preprocessing and feature extraction. However, this approach not only reduces generation efficiency but also heavily relies on the effectiveness and accuracy of the feature extraction methods for the quality of generated data. Directly learning from raw data and generating desired outcomes has thus emerged as an attractive research direction. This approach aims to minimize intermediate processing steps, thereby avoiding potential information loss, and it aligns more closely with the end-to-end deep learning trend.

EEG-GAN [57] is the first work to apply the GAN framework to generate raw EEG data. To apply the GAN to generate naturalistic samples of EEG data, an improvement is proposed to the Wasserstein GAN training showing increased training stability. This research explored and discussed a series of methods and architecture choices for time-series generation networks. Evaluation was conducted using metrics such as inception score and FID distance, showing that the EEG-GAN framework generated naturalistic EEG examples. Sharaj et al. [58] introduced an architecture that combines bilinear upsampling with bilinear weight initialization for deconvolution layers. This two-step upsampling method helps prevent frequency artifacts and stabilize GAN training.

These efforts successfully generated data within the original signal space, demonstrating the feasibility of GANs for this problem. However, these studies were limited to generating single-channel EEG data.

To date, there have been relatively few studies on the generation of multi-channel raw EEG signals, the primary methods employed include CNN-based GANs and RNN-based GANs.

Zuochen Wei et al. [59] directly applied WGAN-GP to generate multi-channel EEG data, demonstrating its effectiveness in enhancing the classification accuracy of epileptic EEG models. Fatemeh F. [42] proposed a framework based on deep convolutional generative adversarial networks (DCGAN) to generate EEG data for augmenting training sets, aiming to improve the performance of brain–computer interface (BCI) classifiers. Sherif M. [36] introduced a cyclic generative adversarial network model incorporating an RNN layer in the generator's network structure. Roy et al. [43] utilized long short-term memory (LSTM) networks within GANs to extract temporal features from EEG data.

CNNs offer advantages in processing spatial features; however, their effectiveness in global features is limited by kernel size. RNNs excel in capturing robust temporal features but struggle to extract effective spatial information. For long sequence EEG signals, RNNs cannot perform parallel computation efficiently, thus leading to lower efficiency.

Table 1 provides a comprehensive analysis of existing method discussed in the literature.

Table 1. Existing methods of EEG data augmentation.

Reference	Dataset	Form of DA	DA Method	Analysis
[20]	SEED	DE	CWGAN	The data undergoes feature extraction before DA; Efficiency of DA reduced; The quality of generated data relies on the accuracy and effectiveness of the feature extraction.
[23]	BCIC IV 1 and 2b	spectrogram images	DCGAN	
[27]	DEAP	descriptive features	WGAN-GP	
[28]	BCIC IV 1 and 2b	topographic maps	FastGAN-ASP	
[29]	Private dataset	one channel EEG data	GAN	Limited to generating single channel EEG data
[30]	BCIT X2	one channel EEG data	CWGAN-GP	
[31]	CHB-MIT Scalp	Raw EEG data	DCGAN	Insufficient ability for CNN to capture global features
[14]	Private dataset	Raw EEG data	DCGAN	
[15]	PhysioNet	Raw EEG data	Recurrent GAN	Insufficient ability for RNN to capture spatial features; Vanishing gradients over long sequences.
[16]	BCIC IV 2b	one channel EEG data	LSTM-GAN	

2.2. Transformer

There remains a need to explore effective ways of integrating and extracting spatiotemporal information from EEG data.

The transformer architecture, first introduced by Vaswani et al. in 2017, revolutionized natural language processing (NLP) with its self-attention mechanism, which excels in processing sequential data. Numerous studies in NLP have demonstrated its performance [49,60].Capitalizing on its robust feature extraction capabilities, researchers have extended transformer models to computer vision tasks. In various visual studies, transformer-based models have achieved performance comparable to or surpassing other networks. For instance, the vision transformer (ViT) [61] applies transformer principles to image classification tasks; Tts-gan [62] employs a transformer-based generative adversarial network for synthetic image generation, leveraging discriminator designs inspired by ViT.

Current research indicates that transformers excel in modeling long-range dependencies and have demonstrated strong performance in both computer vision (CV) and NLP domains. They have been shown to surpass many other popular neural network architectures, such as CNNs for images processing and RNNs for sequential data analysis. These findings highlight transformers' versatile capabilities [63] and suggest promising avenues for their application in EEG data generation and analysis.

3. Materials and Methods

3.1. WGAN-GP Framework

Generative adversarial networks (GANs) consist of two fundamental components: the generator and the discriminator. The generator initiates by sampling from a low-dimensional distribution and maps these samples to a high-dimensional space where real data resides. Subsequently, the discriminator assesses the output generated by the generator. Through adversarial training, both components iteratively improve their respective generation and discrimination capabilities, facilitating the generator to produce synthetic data similar to the original data. This study adopted the WGAN-GP architecture, which stabilizes model training through a gradient penalty strategy. The framework of WGAN-GP is illustrated in Figure 1.

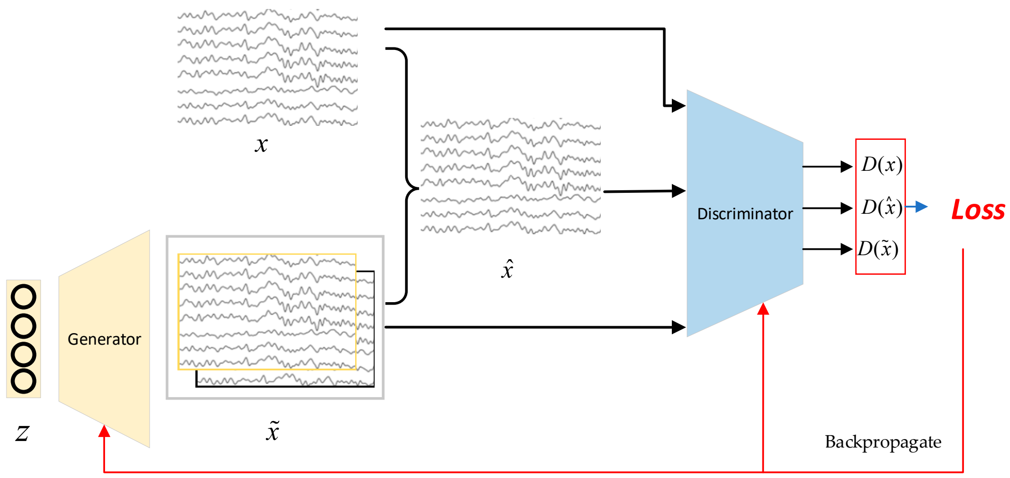


Figure 1. WGAN-GP architecture.

Here, z is sampled from Gaussian noise, x represents the original data, and $\tilde{x}$ represents the generated data. During the training process, a significant distribution gap exists between the generated and original data, requiring multiple optimization iterations for the GAN to converge. Given the limited training data, the discriminator is prone to overfitting, which in turn hinders its ability to provide effective gradients for the generator. To mitigate the impact of data scarcity, instead of directly fitting the distribution of the original data, we incorporate the generator's output g(z) into the original data. Therefore, the calculation of generated data is as shown in Equation (1).

$$\widetilde{x} = \mathrm{g}(z) + x. \tag{1}$$

This approach reduces the distribution gap between the original data and the generated data compared to directly fitting the generator to the original data distribution. It accelerates model convergence and prevents the generator from training halts caused by discriminator overfitting. $\hat{x}$ represents a stochastic mixture of real and generated data, used to calculate the gradient penalty terms to stabilize the training process. The calculation of $\hat{x}$ is shown in Equation (2); ε is a random number uniformly distributed between (0, 1).

$$\hat{x} = \varepsilon x + (1 - \varepsilon)\widetilde{x}. \tag{2}$$

The loss function is defined as shown in Equation (3), where λ is an adjustable hyperparameter that represents the weight of the gradient penalty term.

$$L = \underset{\widetilde{\mathbf{x}} \sim \mathbb{P}_g}{\mathbb{E}}[D(\widetilde{\mathbf{x}})] - \underset{x \sim \mathbb{P}_r}{\mathbb{E}}[D(\mathbf{x})] + \lambda \underset{\hat{\mathbf{x}} \sim \mathbb{P}_{\hat{\mathbf{x}}}}{\mathbb{E}}\left[(\|\nabla_{\hat{\mathbf{x}}} D(\hat{\mathbf{x}})\|_2 - 1)^2\right]. \tag{3}$$

WGAN-GP employs alternating training of the generator and discriminator. The training process is outlined as Algorithm 1:

Algorithm 1: Gradient penalty weight λ = 10, Number of discriminator updates before each generator update n = 3, Learning rate α = 0.000, Batch size m = 128

```
Input: Original data, Gaussian noise
Output: Trained generator and discriminator
while model has not converged do:

    for t = 1 to n do:Sample from original data {x^1, x^2, ..., x^m};
    Sample from Gaussian noise {z^1, z^2, ..., z^m};
        Compute generated data {x̃^1, x̃^2, ..., x̃^2}
        Compute loss function L
        Backpropagate and update discriminator parameters
    end for
    Sample from Gaussian noise {z^1, z^2, ..., z^m}
    Compute generated data {x̃^1, x̃^2, ..., x̃^2}
    Compute loss function L
    Backpropagate and update generator parameters
end while
```

3.2. Generator Network Architecture

The generator is primarily composed of upsampling layers and convolutional layers. Upsampling layers employ bilinear interpolation to double the size of the input, while convolutional layers decrease the number of output channels and adjust the size. The structure of the generator is detailed in Table 2.

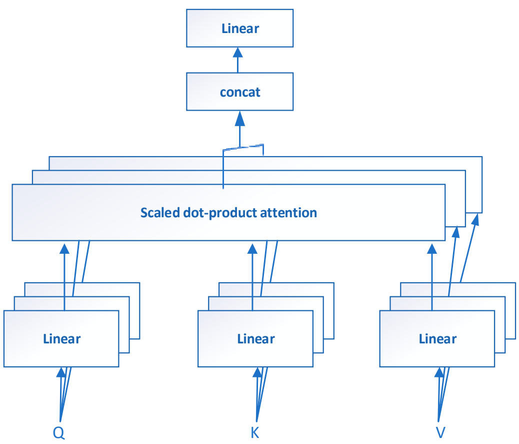
The generator takes as input a 100-dimensional noise vector sampled from a standard normal distribution. This noise vector undergoes a linear transformation to map it to a size of 64 × 4 × 4. Subsequently, there are three upsampling operations using bilinear interpolation followed by convolutional layers with 5 × 5 kernels. Finally, the output passes through a 1 × 15 convolutional layer and a 22 × 1 convolutional layer to adjust the size to match the original data. Batch normalization layers are used for regularization throughout the process, and ReLU activation functions are applied except for the final layer, which uses tanh activation to map the output values into the range [−1, 1].

Table 2. Generator network architecture.

Layer	Size	Output Shape	Activation
input		[1, 100]	
linear	$64 \times 4 \times 4$	[1, $64 \times 4 \times 4$]	
Upsample		[64, 8, 8]	
Conv2D	5×5	[32, 8, 8]	relu
Upsample		[32, 16, 16]	
Conv2D	5×5	[16, 16, 16]	
Batchnorm		[16, 16, 16]	relu
Upsample		[16, 32, 32]	
Conv2D	5×5	[8, 32, 32]	
Batchnorm		[8, 32, 32]	relu
Upsample		[8, 64, 64]	
Conv2D	1×15	[1, 64, 50]	
Batchnorm		[1, 64, 50]	relu
Conv2D	22×1	[1, 22, 50]	tanh

3.3. Discriminator Based on Spatiotemporal Features

Due to the limitations of CNNs and RNNs in generating EEG data, and the demonstrated versatility of transformer networks in extracting temporal and spatial features, this paper proposes an EEG data augmentation model based on the transformer encoder structure, the discriminator is designed to extract both temporal and spatial features simultaneously, employing scaled dot-product attention mechanism shown in Figure 2a and multi-head attention mechanisms shown in Figure 2b.



(**a**) (**b**)

Figure 2. (**a**) Multi-head attention; (**b**) scaled dot-product attention.

Attention function is a mathematical operation in which the input consists of a set of queries and a set of key-value pairs. It calculates attention weights for the values based on queries Q and keys K, and then calculates the weighted sum of values V to produce the output of the attention function. The computation of scaled dot-product attention can be expressed by Equation (4):

$$\text{Attention}\ (Q, K, V) = \text{softmax}\left(\frac{QK^{\top}}{\sqrt{d_k}}\right)V, \tag{4}$$

where Q represents the query matrix, K represents the key matrix, V represents the value matrix, derived from the input after positional encoding and linear projection, dk represents the dimensionality of the projected vectors. QK^T denotes the dot product of Q and the transpose of K, $\sqrt{d_k}$ is a scaling factor aimed at stabilizing gradients, and softmax is applied row-wise to compute attention weights. This formula effectively computes how much each value should contribute to the output based on the compatibility between queries and keys.

Q, K, and V matrices are linearly projected with different, learned linear projections into d_k dimensions and attention functions are then executed in parallel. This multi-head attention enables the model to jointly attend to information from different representation subspaces at different positions (5):

$$\begin{array}{c} \text{MultiHead}(X) = \text{Concat}(\text{head}_1, \ldots, \text{head}_\text{h})W^O \\ \text{where head}_\text{i} = \text{Attention}\left(XW_i^Q, XW_i^K, XW_i^V\right) \end{array} . \tag{5}$$

The discriminator analyzes the acquired raw and generated data, structured as shown in Figure 3. Since this model does not include convolutional or recursive structures. To enable the discriminator to extract sequential features in time, and dependencies between different channels, positional encoding of the raw data is necessary. Firstly, input data are positionally encoded using a set of learnable parameters. Subsequently, temporal and spatial autoencoders are employed to extract temporal and spatial features of the sequence data. To mitigate sequence-related biases, the temporal and spatial autoencoders operate in parallel. Finally, a linear layer integrates the spatiotemporal features to compute the output values.

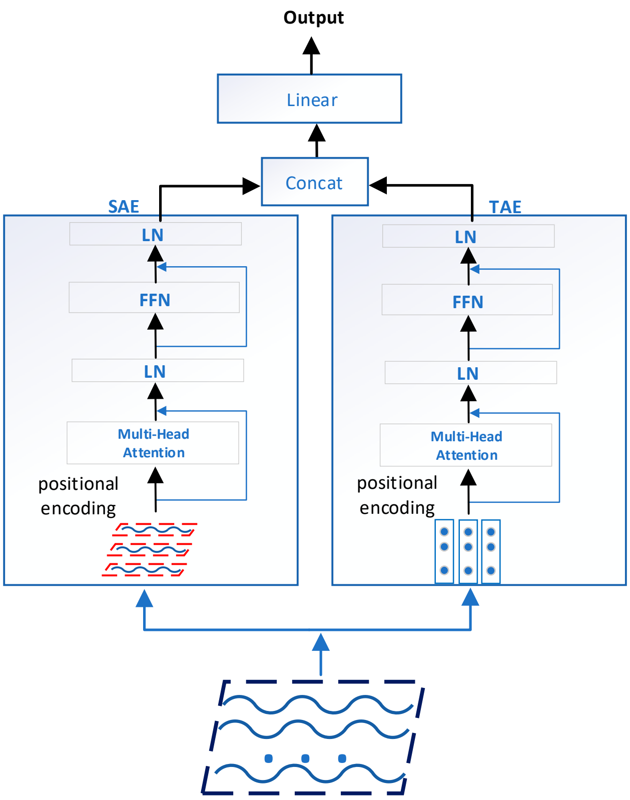


Figure 3. STGAN architecture.

This paper employs two methods to compute attention. The input format of the discriminator is represented as a matrix of size C × T, where C denotes the number of channels, and T represents the number of sampling points per channel

Different channels correspond to different sampling positions, each reflecting specific regions of the brain. In order to extract spatial features that reflect the functional connections between these regions, we designed the spatial attention extraction (SAE) module. The module takes as input the matrix $[s_1, s_2, \ldots, s_C](s_i \in R^T)$, positional encoding of the sampled data from each channel, followed by the calculation of a self-attention function. This process captures spatial correlations and extracts features. The computation proceeds as follows.

Perform spatial positional encoding as Equation (6). Here, S_0 denotes the input to SAE, and E_{pos} is a set of learnable positional parameters match the same size as S_0:

$$S_1 = S_0 + E_{pos}, E_{pos} \in R^{C \times T}. \tag{6}$$

Compute multi-head self-attention and apply residual connection and layer normalization as Equation (7):

$$S_2 = LN(MultiHead(S_1) + S_1). \tag{7}$$

The feedforward network (FFN) layer consists of two linear transformations with a nonlinear activation function, ReLU, between them, as Equation (8). The output of the feedforward network is computed, and residual connection and layer normalization are applied, as in Equation (9):

$$FFN(X) = Linear2(Relu(Linear1(X))). \tag{8}$$

$$S_3 = LN(FFN(S_2) + S_2). \tag{9}$$

In the time encoding section, we designed a temporal attention extraction (TAE) module to compute the correlations between sampling points. As illustrated in the figure, this module transposes the input data into a matrix $[t_1, t_2, \ldots, t_T], t_i \in R^C$, applies positional encoding to the data at each time step, and then calculates the multi-head attention to extract temporal features of EEG. The calculation process is as follows.

Firstly, transpose the input matrix, as shown in Equation (10).

$$T_0 = trans(input), T_0 \in R^{T \times C}. \tag{10}$$

Then, positional encoding and feature extraction are performed, as described in Equations (11) to (13).

$$T_1 = T_0 + E_{pos}, E_{pos} \in R^{T \times C}. \tag{11}$$

$$T_2 = LN(MultiHead(T_1) + T_1). \tag{12}$$

$$T_3 = LN(FFN(T_2) + T_2). \tag{13}$$

Finally, the global information is aggregated, and the classification results are outputted. The discriminator structure is depicted in Table 3.

Table 3. Discriminator network architecture.

Block	Layer	Output Shape	Activation
SAE	input	[1, 22, 50]	
	Positional encoding	[1, 22, 50]	
	Multi-Head Attention	[1, 22, 50]	relu
	Dropout	[1, 22, 50]	
	layernorm	[1, 22, 50]	
	Feed-Forward Network	[1, 22, 50]	relu
TAE	transpose	[1, 50, 22]	
	Positional encoding	[1, 50, 22]	
	MultiHead	[1, 50, 22]	relu
	Dropout	[1, 50, 22]	
	Layernorm	[1, 50, 22]	
	Feed-Forward Network	[1, 50, 22]	relu
	Dropout	[1, 50, 22]	
	LayerNorm	[1, 50, 22]	
	transpose	[1, 22, 50]	
	concat	[1, 44, 50]	
classfier	FC	[1, 44, 1]	relu
	FC	1	

The discriminator accepts original data and generated data of size [1, 22, 50]. It processes them through parallel operations of the SAE module and the TAE module to extract spatial and temporal features, respectively. The SAE module receives inputs of size [1, 22, 50] and begins with positional encoding. It includes a multi-head attention layer with four fully connected layers, each containing 50 neurons, two fully connected layers in the feedforward network with neuron counts of 100 and 50, respectively, uses ReLU activation functions and layer normalization. The TAE module initially transposes the input dimensions to [1, 50, 22] and follows a structure similar to SAE. The key difference lies in the number of neurons in each fully connected layer. Finally, the outputs from SAE and TAE are concatenated and processed through two additional fully connected layers to obtain the final output.

4. Experiments

4.1. Dataset and Preprocessing

This study employs EEG signals elicited during motor imagery (MI) for identity recognition. Motor imagery refers to a specific cognitive process where subjects mentally simulate actions they intend to perform. During this process, even without physically executing the actions, similar brain regions to those involved in actual movement are activated, generating distinct EEG signals. Motor imagery tasks are particularly well suited for brainprint identification because they involve conscious, voluntary mental simulation from subjects. EEG signals elicited during motor imagery require subjects to be alert and willing without the need for physical movement, thereby ensuring discretion.

The dataset used in this study is BCI Competition IV Dataset 2A [64]. This dataset records EEG data during motor imagery tasks involving left hand, right hand, both feet, and tongue movements performed by 9 subjects. Each subject performed 72 trials of each of the 4 tasks during a single experiment, and each motor imagery trial lasted for 3 s. The EEG data were recorded using 22 Ag/AgCl electrodes at a sampling frequency of 250 Hz and were bandpass filtered between 0.5 and 100 Hz.

Table 4 summarizes the information of the dataset, including the EEG channel list involved in the study. EEG electrode montage is shown in Figure 4, corresponding to the international 10–20 system.

Table 4. Dataset information.

Task	Subject	Channel	EEG Channel List
MI	9	22	Fz, FCz, FC1, FC2, FC3, FC4, Cz, C1, C2, C3, C4, C5, C6, CPz, CP1, CP2, CP3, CP4, Pz, P1, P2, POz

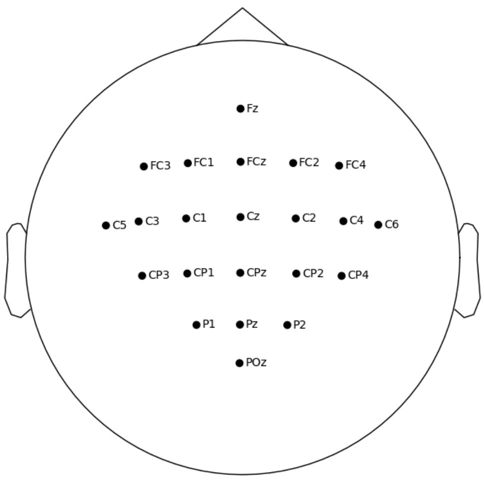


Figure 4. EEG electrode montage.

Data augmentation and identity recognition were conducted under consistent individual states, using only EEG data from left-hand motor imagery tasks in this paper. The Python package MNE was used for preprocessing the EEG data. An additional 50 Hz notch filter was enabled to suppress line noise; three EOG channels recording eye movements were excluded. For each individual's EEG data, a third-order Butterworth IIR filter was applied in the 4–40 Hz frequency band to reduce the influence of eye movements. A fifth-order Butterworth IIR bandpass filter with a frequency range of 4 Hz to 40 Hz was applied to the raw data object. A sliding window of size 50 was used to segment the data, corresponding to a sample duration of 0.2 s, with a sliding step of 50 to ensure no overlap between samples. Subsequently, the data were normalized using min–max normalization to the range [0, 1], and then multiplied by 2 and subtracted by 0.5, resulting in a range of [-1, 1]. The dataset was divided into training and testing sets in a 4:1 ratio, with each individual's training set consisting of 864 samples.

4.2. Baseline Methods for EEG Generation

This paper introduces a novel EEG data augmentation method based on transformers. At present, the generation of EEG data based on GAN mostly adopts convolutional or recurrent neural network structures. Therefore, we implemented two other competitive EEG data augmentation methods as benchmarks: RGAN-based EEG data augmentation [36] and DCGAN-based EEG data augmentation [42]. Additionally, to evaluate the performance of different network architectures in the EEG data augmentation task, we also implemented a basic fully connected GAN as a baseline.

The architecture of the baseline RGAN model is specifically designed to capture temporal dependencies in signals and generate realistic signal samples. It consists of two parts: a generator and a discriminator. The generator in RGAN is composed of four hidden layers, the first two layers are RNNs designed to capture time dependencies in the signals each with 256 neurons, while the other two are fully connected (FC) layers with 128 and 50 neurons, respectively. It outputs a generated signal sample that works as an input of the discriminator component. The discriminator has three hidden fully connected layers with 256, 128, and 1 neurons, respectively.

In the DCGAN, the generator network commences with two fully connected layers, comprising 256 and 400 neurons, respectively, followed by a batch normalization layer. The output is then reshaped to dimensions of (10, 40) and subsequently upsampled using bilinear interpolation with a scale factor of 5. This is followed by a convolutional layer featuring a kernel size of 3×3, a stride of 2, and padding of 1, with input and output channels of 1 and 64, respectively. The next stage involves another convolutional layer with a kernel size of 3×3, a stride of 1, and padding of (0, 1), where the input and output channels are 64 and 1, respectively. This configuration ensures that the output of the generator aligns with the shape of the original EEG signals. The discriminator begins with a convolutional layer characterized by a kernel size of 3×3, a stride of 2, and padding of 1, with input and output channels of 1 and 64, respectively. This is followed by a max pooling layer of size 2. Subsequently, the output is processed through another convolutional layer, which also has a kernel size of 3×3, a stride of 2, and padding of 1, with input and output channels of 64 and 1, respectively. Another max pooling layer of size 2 is applied, then flattening the output to dimensions of (1, 44), and finally a fully connected layer of size 1. Tanh activation function is used in both networks except in the last layer of the discriminator.

Both the generator and discriminator are of GAN composed of fully connected neural networks. The generator consists of three fully connected layers: the first two layers contain 256 and 550 neurons, respectively, after which the output is reshaped to dimensions of [25, 22]. The final fully connected layer comprises 50 neurons, ensuring that the generator's output size matches that of the original data. The discriminator consists of four fully connected layers. The first layer contains 25 neurons, with its output being flattened. The subsequent three fully connected layers contain 256, 100, and 1 neuron, respectively. The connections between the fully connected layers are facilitated by ReLU activation functions and batch normalization layers.

To train the generative models, ADAM optimizer is used, with a beta1 parameter set to 0.2, the learning rate is 0.0001, and the batch size is 128.

4.3. Evaluation Metrics

(1) FID is a widely used metric for evaluating generative models. It measures the distance between feature vectors of real and generated samples, aiming to assess the quality of generated samples more effectively. FID utilizes a pre-trained classifier to compare the feature representations of real and generated samples at an intermediate layer. Heusel et al. [50] have demonstrated that FID correlates well with human judgment of sample quality, indicating its effectiveness in assessing the realism of generated samples. FID exhibits strong robustness to noise in inputs, providing informative insights into the quality of generated samples while also being sensitive to mode collapse issues. The calculation of FID is shown in Equation (14)

$$FID = \left\|\mu_r - \mu_g\right\|^2 + \mathrm{Tr}\left(\Sigma_r + \Sigma_g - 2\left(\Sigma_r\Sigma_g\right)^{\frac{1}{2}}\right) \tag{14}$$

(2) Visual Evaluation: Principal component analysis (PCA) is a data dimensionality reduction algorithm that represents the distribution of data in a lower-dimensional space by extracting principal component vectors. In EEG analysis, PCA is not only used for dimensionality reduction but also for matching data rank with the data dimension [65]. Given to the intrinsic complexity of EEG data, which is not easily interpretable, we employ PCA to reduce the dimensionality of real and generated data. This allows us to visually compare and assess both types of data by mapping them onto a two-dimensional space.

(3) To validate the effectiveness of the data we generated in improving recognition accuracy, we trained EEGNet [66], DeepConvNet [52], and ShallowConvNet [52] models on both the original training set and the augmented training set. These deep learning models are widely used for EEG classification tasks.

EEGNet is a compact convolutional neural network designed specifically designed for EEG classification tasks. It has achieved success across multiple EEG paradigms. The model architecture includes three convolutional layers designed to learn temporal and spatial filters, which are crucial for processing EEG data. These convolutional layers are followed by a fully connected layer responsible for classification tasks, making EEGNet effective in extracting and leveraging intricate patterns present in EEG signals.

DeepConvNet is another general architecture composed of five convolutional layers followed by a softmax layer for classification purposes.

ShallowConvNet is specifically designed for oscillatory signals, achieving classification by extracting features related to logarithmic band powers. It consists of two convolutional layers, a nonlinear square layer, an average pooling layer, and a nonlinear logarithm layer.

We implemented these models using PyTorch, trained them with a learning rate of 0.001, Adam optimizer, and a batch size of 512.

To evaluate the validity of the experimental results, we conduct statistical tests. Statistical tests commonly used in deep learning include the *t*-test [67], Kolmogorov–Smirnov-based test [68] and Wilcoxon signed-rank test [69]. In this paper, we used the Wilcoxon signed-rank test to calculate the *p*-values, as performed at baseline [42]. This is because our data are non-normally distributed and involve paired comparisons, making the Wilcoxon signed rank test a more appropriate choice.

4.4. Results and Analysis

4.4.1. Fid Results and Analysis

This paper trained an EEGNet classifier on the training set of real data. Features of real and generated samples were obtained for each individual before the classification layer, and then the FID between them was calculated. A smaller FID indicates a smaller distribution difference between generated and real samples. The computed results are shown in Table 5.

Table 5. FID scores of data generated by different methods.

Subject	GAN	DCGAN	RGAN	STGAN
1	44,928	37.403	41.006	**34.962**
2	52,034	56.856	**46.789**	56.213
3	128,191	17.314	18.302	**16.687**
4	125,873	26.828	32.040	**24.792**
5	52,640	27.312	28.0361	**23.315**
6	81,495	50.143	52.175	**44.876**
7	130,116	94.995	64.574	**53.238**
8	135,086	17.728	19.207	**16.415**
9	89,137	47.579	52.151	**47.146**
Mean	44,928	41.795	45.281	**35.294**

The values in bold represent the best results.

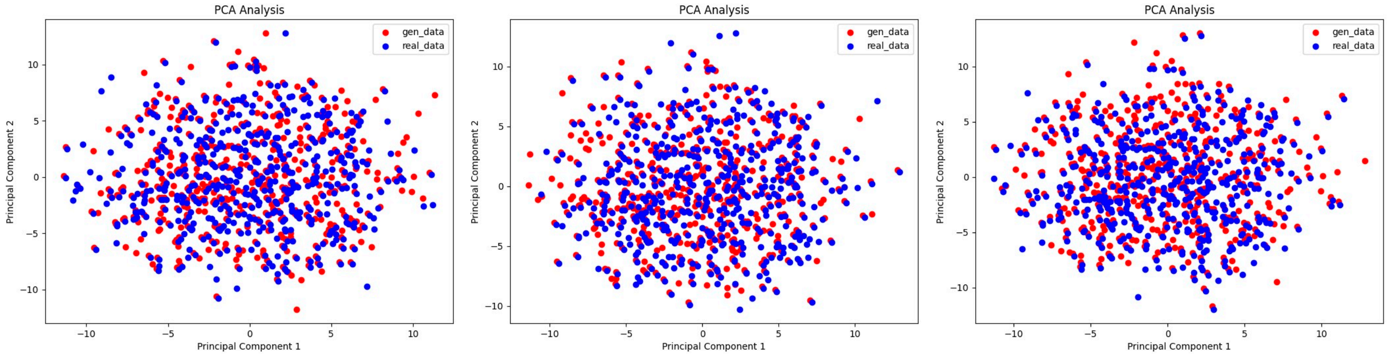
According to the results in Table 4, GAN based on fully connected neural network is not an effective EEG data augmentation method. The FID of the data generated by the GAN is significantly higher than that of the data generated by DCGAN, RGAN, and STGAN, indicating that the data generated by GAN differ greatly from the original data in the feature space. This suggests that the fully connected neural network cannot effectively extract features from EEG data, resulting in the inability to generate EEG data that have similar features to the original data.

Even using the same generative model, there are variations in FID scores between generated and original data across different individuals, highlighting the individual differences of EEG data. Consequently, capturing features of the EEG data from different individuals and effectively performing data augmentation pose varying levels of complexity. Notably, for individual 2, RGAN outperformed DCGAN and STGAN, indicating the significant contribution of temporal features to data generation quality in this individual. Due to the increased model complexity compared to RGAN, STGAN did not achieve optimal performance in extracting temporal features. The FID scores for individual 7 are significantly higher compared to others, indicating greater difficulty in argumenting EEG data. This difficulty may arise from factors such as lower-quality original data, less distinct features, or limitations in the generative model's capability to extract features from the data.

More significantly, the FID scores of the data generated by STGAN surpass those of DCGAN ($p < 0.02$) and RGAN ($p < 0.05$), with statistically significant differences observed. While with some individuals, such as individual 3, FID scores show no significant differences, indicating comparable generation capabilities among the three models, across all individuals, the average FID was reduced by 7.015 compared to DCGAN and by 9.987 compared to RGAN, indicating that the generated data by STGAN were of higher quality. This is due to the encoder structure and the self-attention mechanism of the transformer overcoming the deficiencies in global feature extraction of CNNs and spatial feature extraction of RNNs, allowing for a more comprehensive and accurate capture of temporal and spatial features in EEG data. As a result, compared to the generated data by DCGAN and RGAN, the generated data by STGAN are closer to the real data in the feature space, especially in individual 7, where STGAN reduced the FID by 41.805 compared to DCGAN and by 11.336 compared to RGAN. This indicates that even in challenging situations, the generation capability of STGAN remains superior to that of DCGAN and RGAN.

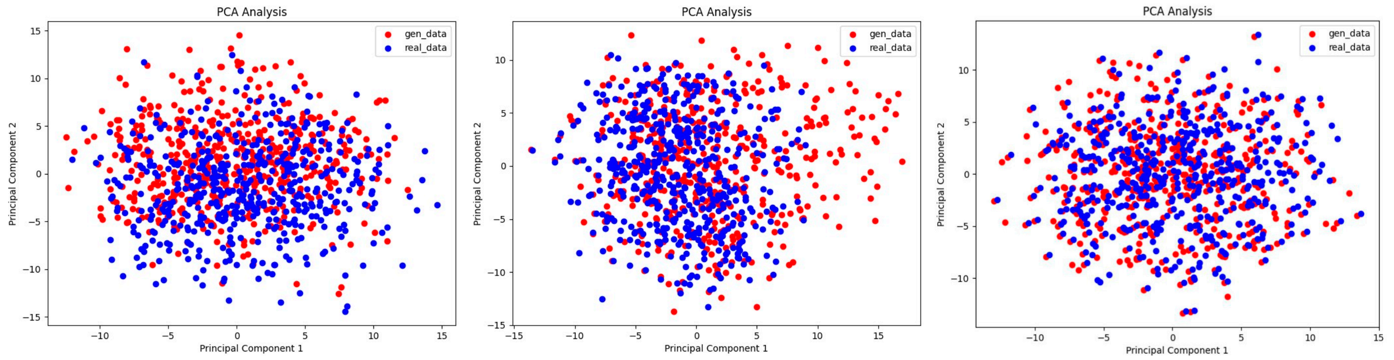
4.4.2. Visual Evaluation

The PCA algorithm is employed to project both the original and generated data into a two-dimensional space for visualization. This algorithm provides an intuitive demonstration of the qualitative similarity between the distributions of generated and original samples in reduced-dimension space. Closer distributions indicate greater similarity, suggesting the higher fidelity of the generated samples to the original data. The data generated by STGAN for individual 3 and individual 7 have the lowest and highest FID scores, respectively; Figures 5 and 6 show the PCA visualization results of them.



(**a**) (**b**) (**c**)

Figure 5. (**a**) Original data of individual 3 and data generated by DCGAN; (**b**) original data of individual 3 and data generated by RGAN; (**c**) original data of individual 3 and data generated by STGAN.



(**a**) (**b**) (**c**)

Figure 6. (**a**) Original data of individual 7 and data generated by DCGAN; (**b**) original data of individual 7 and data generated by RGAN; (**c**) original data of individual 7 and data generated by STGAN.

Generated data from DCGAN, RGAN, and STGAN are all relatively close to the original data for individual 3, indicating better generation capabilities across all three models for this individual. In contrast, for individual 7, there are noticeable differences between the generated data distributions of the three models and the original data, highlighting the difficulty in generating data for this individual. However, compared to the other two models, STGAN exhibits a generated data distribution closer to the original data distribution on individual 7. This suggests that even under challenging generation conditions, STGAN can still achieve better generation performance than RGAN and DCGAN, indicating stronger feature extraction capabilities consistent with the FID analysis results.

4.4.3. Classification Accuracy Improvement

In order to evaluate the impact of data augmentation methods on classification models, this paper implemented DCGAN, RGAN, and STGAN for EEG data augmentation. For comparison, EEGNET, ShallowConvNet and DeepConvNet were trained on the datasets before and after data augmentation. The recognition accuracy of each individual is shown in Figures 7–9, and the results indicate that for EEGNET, ShallowConvNet and DeepConvNet, STGAN enhancement achieved the highest accuracy, whereas the performance of the three data argumentation methods was comparable on DeepNet.

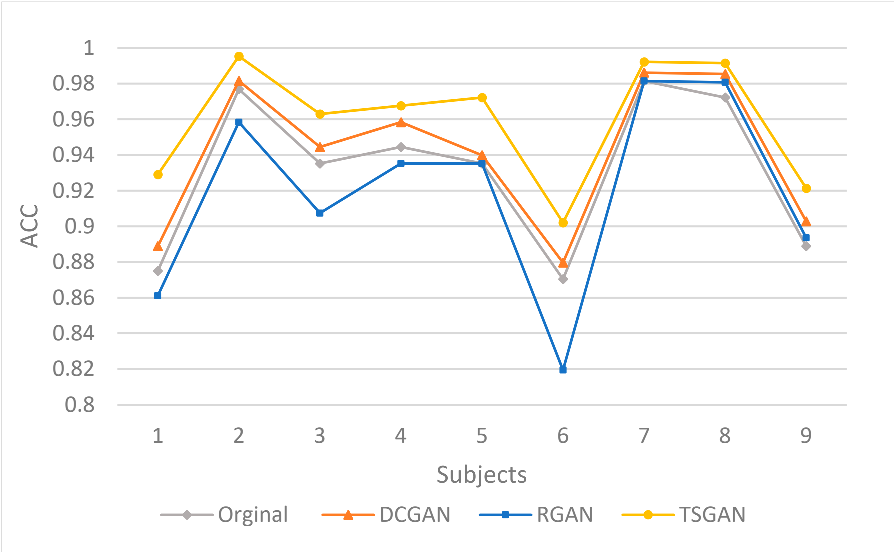


Figure 7. Individual recognition accuracy of EEGNET.

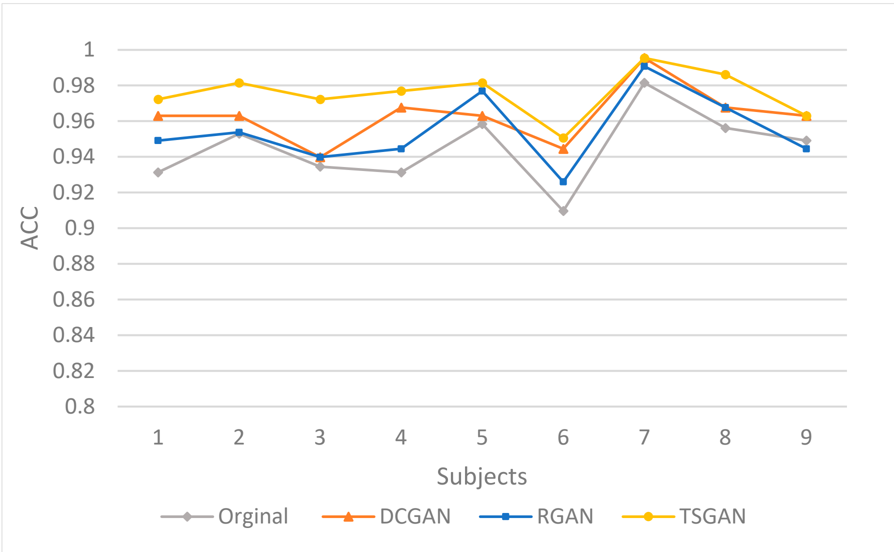


Figure 8. Individual recognition accuracy of ShallowConvNet.

The average recognition accuracy of all individuals is shown in Table 6. On the dataset used in this paper, RGAN-generated data improved the accuracy of ShallowConvNet and DeepConvNet but negatively affected the classification accuracy of EEGNET. In contrast, both DCGAN and STGAN effectively enhanced the classification performance across all three models, showcasing superior generalization compared to RGAN. Specifically, DCGAN increased the accuracy of EEGNET, ShallowConvNet, and DeepConvNet by 1.1% ($p < 0.005$), 2.13% ($p < 0.005$), and 1.69% ($p < 0.05$), respectively, while STGAN improved the accuracy of these networks by 2.49% ($p < 0.005$), 2.59% ($p < 0.005$), and 1.14% ($p < 0.02$), respectively; the results are statistically significant.

The experimental outcomes suggest that STGAN serves as an effective approach for data augmentation, showcasing a wider applicability across diverse EEG classification models and achieving more pronounced improvements in performance over alternative methods. This enhanced performance is credited to the spatiotemporal feature-based discriminator within the STGAN, which ensures that the generated data maintain latent features similar to the original dataset, consequently minimizing the distributional discrepancies between synthetic and authentic data. Augmenting EEG training dataset with

STGAN helps models learn more robust and generalized feature representations, thereby improving their classification performance.

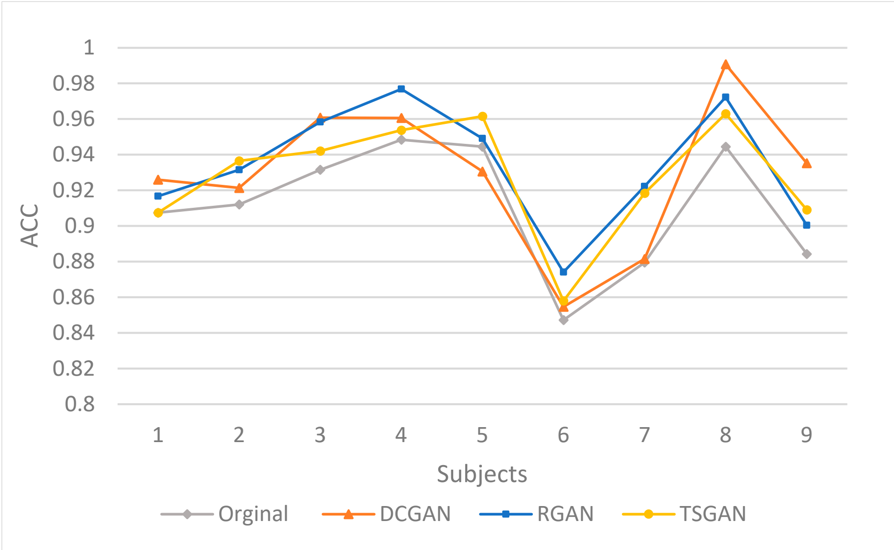


Figure 9. Individual recognition accuracy of DeepConvNet.

Table 6. The average recognition accuracy of all individuals.

	Original	DCGAN	RGAN	GAN
eegnet	93.79	94.90	92.52	**96.28**
Shallownet	94.80	96.93	95.24	**97.39**
Deep4net	90.93	92.07	92.33	**92.62**

The values in bold represent the best results.

4.5. Ablation Study

STGAN simultaneously extracts temporal and spatial features through the SAE and TAE modules, respectively. To further elucidate how the SAE and TAE modules individually impact the quality of generated data and identity recognition performance, we conducted an ablation study on STGAN. By removing the SAE and TAE modules separately, we evaluated the generated data using the FID metric and conducted identity recognition experiments. This allowed us to observe and analyze their contributions to the overall performance.

In Table 7, the FID results show that the GAN utilizing both the SAE and TAE modules generates data with better FID scores than those using only the SAE ($p < 0.005$) or TAE ($p < 0.005$). This finding highlights the importance of simultaneously extracting temporal and spatial features for EEG data generation, leading to higher data quality compared to methods that focus on a single feature type.

Table 7. FID scores of generated data in ablation study.

Subject	SAE	TAE	SAE + TAE
1	38.263	35.688	34.962
2	63.665	65.226	56.213
3	18.232	17.106	16.687
4	26.292	28.486	24.792
5	27.075	27.347	23.315
6	50.642	51.228	44.876
7	72.9	55.880	53.238
8	16.958	19.376	16.415
9	51.358	50.411	47.146
MEAN	40.598	38.972	35.294

As shown in Table 8, when only SAE is used for data augmentation, the accuracy of the three classification models is lower than that both SAE and TAE are used ($p < 0.02$). This indicates that although spatial features are important for identity recognition based on EEG data, relying solely on spatial features is not sufficient to achieve optimal performance. When only TAE is used for data augmentation, the accuracy of the classifiers is also lower than that both SAE and TAE are used ($p < 0.02$), but higher than that of the SAE ($p < 0.05$). This suggests that temporal features also make a significant contribution to the identity recognition and may be more critical than spatial features. When both SAE and TAE are included, the accuracy of the three classifiers all reaches the highest level. This demonstrates that simultaneously extracting spatial and temporal features can significantly improve the quality of the generated EEG data, thereby enhancing the performance of subsequent identity recognition task.

Table 8. The average recognition accuracy of all individuals in ablation study.

	SAE	**TAE**	**SAE+TAE**
EEGNET	90.13	95.60	**96.28**
ShallowConvNet	94.36	95.53	**97.39**
DeepConvNet	91.13	92.11	**92.62**

The values in bold represent the best results.

It can be seen from this that for human identity recognition, temporal information and spatial information are complementary, and temporal information is more important than spatial information. In order to simultaneously acquire the spatio-temporal information of EEG, our model is composed of a SAE module and a TAE module, which yielded the best performance of recognition accuracy after data augmentation.

5. Discussion

This paper proposed a novel approach to augment EEG data using generative adversarial networks (GANs), incorporating a temporal attention encoder and a spatial attention encoder to capture global dependencies across channels and time within EEG signals. Key elements of the encoders include learnable positional encoding and self-attention mechanisms, which are designed to improve the quantity of generated EEG data. The WGAN-GP framework is employed to stabilize the training of GAN. By integrating the generator's outputs into the original data, this method reduces the distribution gap between the original and generated data, accelerates model convergence during training and helps prevent the discriminator from overfitting to the generated samples.

Comparative evaluations of STGAN, RGAN, DCGAN and GAN are conducted using PCA and FID. Additionally, the effectiveness of the data augmentation methods is validated by training three classification models on the augmented dataset for identity recognition. The results demonstrate that the EEG data generated by STGAN closely resemble the original data in terms of statistical properties and distribution, showing significant improvements in model accuracy compared to those trained solely on the original dataset. STGAN outperforms DCGAN and RGAN, indicating its ability to overcome the shortcomings of convolutional neural networks and recurrent neural networks in EEG data augmentation. Thus, STGAN proves to be an effective method for addressing the scarcity of EEG data. Ablation studies show that extracting only a single type of feature is insufficient for achieving the highest accuracy of brainprint recognition models, highlighting the necessity of simultaneously extracting temporal and spatial features. Furthermore, we found that extracting temporal features is more effective than spatial features, suggesting its importance in brainprint recognition. However, this study did not include features in frequency domain, which will become our future research direction.

6. Conclusions

EEG data offer unique advantages over traditional biometric features, making them valuable for identity recognition systems, but the scarcity of data limits their development and application. Data augmentation techniques present an effective solution; however, the complex temporal and spatial feature of EEG signals pose challenges for it. This paper proposed an efficient EEG method of data augmentation using STGAN, and demonstrated its superior performance through detailed comparisons and ablation experiments.

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